

University of New Haven



RULEBOOK: RAG-BASED Q&A SYSTEM FOR CRICKET'S LAWS & LLM COMPARISON

DSCI 6004 – Natural Language Processing

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Project Overview

- Build a Retrieval-Augmented Generation (RAG) system
- Use **official ICC Men's Standard ODI Playing Conditions (2023)** as the knowledge base
- Answer cricket rule queries with accuracy & context grounding
- Compare **three open-source LLMs** using the same RAG pipeline
- Evaluate accuracy, relevancy, retrieval quality

Dataset / Knowledge Base

Source: ICC Men's Standard ODI Playing Conditions PDF (Effective December 2023)

Length: ~99 pages of dense rules

Why this domain?

- Hard to search manually
- Rules are very specific & situational
- Ideal for retrieval-based answering

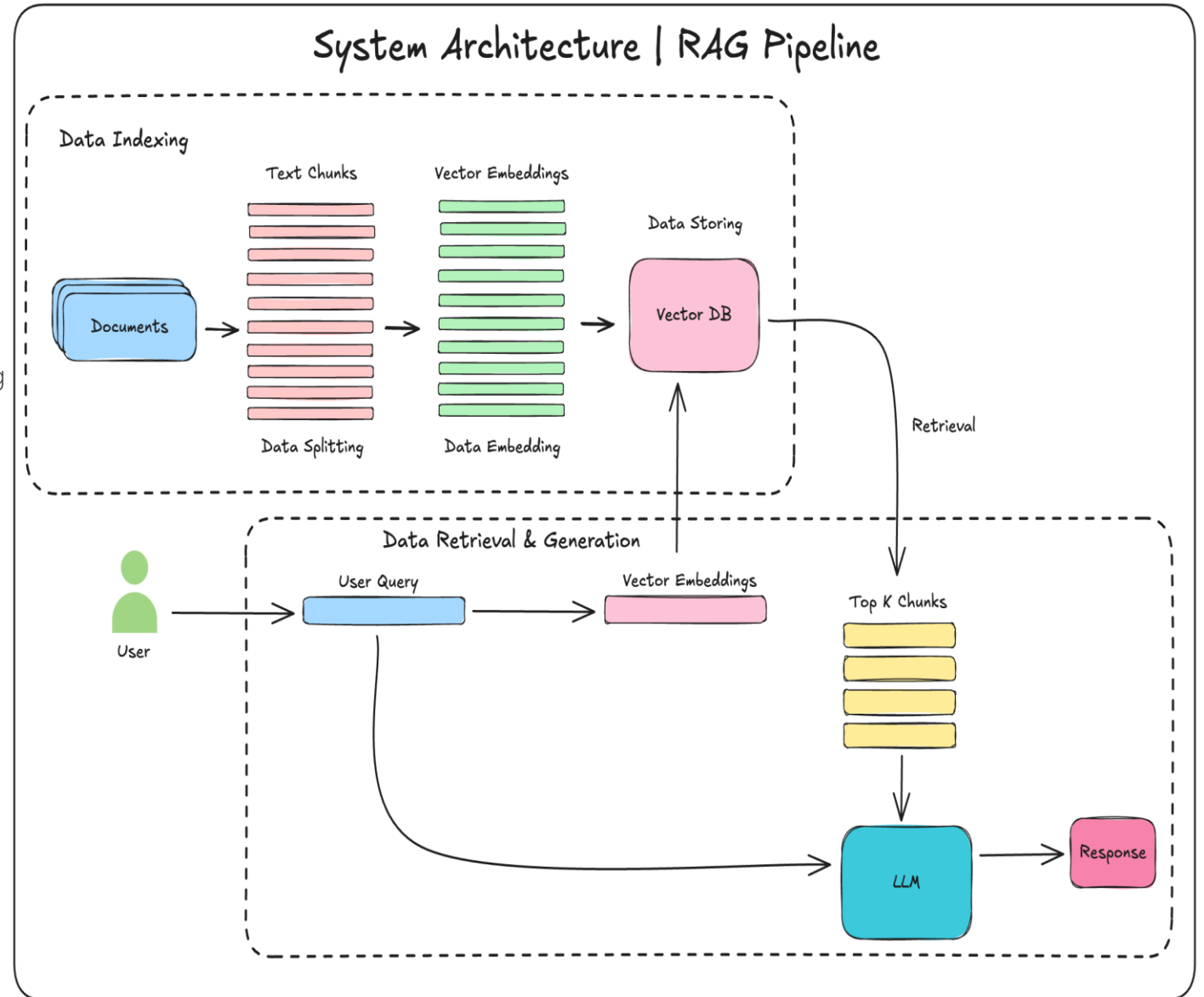
Preprocessing:

- Text extraction
- Chunking into ~1200 token chunks, with 100 token overlapping
- Stored as embeddings in ChromaDB

System Architecture

Pipeline Explanation:

- Document → Split into Chunks
 - The PDF of ICC Men's Standard ODI Playing Conditions is broken into smaller text chunks (1200 tokens).
 - This ensures relevant parts can be retrieved efficiently.
- Chunk Embeddings → Stored in Vector Database (ChromaDB)
 - Each chunk is converted into numeric embeddings using **all-MiniLM** via Ollama.
 - Stored inside **ChromaDB** for similarity search.
- User Query → Embedded & Compared to Stored Chunks
 - The query is embedded the same way as chunks.
 - The system finds the most similar chunks using **cosine similarity**.
- Relevant Chunks → Sent to the LLM
 - The top-k retrieved chunks are passed as context to the selected LLM.
- LLM Generates Output
 - The LLM (Llama 3.2 / Model B / Model C) produces a rule-based answer.
 - Output is grounded in the official cricket laws.



The Three LLMs Used

1. Llama 3.2

- Model Name: llama3.2:latest (~2.0 GB)
- Strong reasoning
- Best contextual grounding
- Most accurate answers

2. Phi-3

- Model Name: phi3:latest (~2.2 GB)
- Balanced quality and speed
- Reliable for medium complexity questions

3. Mistral

- Model Name: mistral:latest (~4.4 GB)
- Lightweight, fast inference
- Lowest computation cost

4. Embedding Model

- Model Name: all-minilm:latest (~45 MB)
- Used for vector embeddings (ChromaDB)
- Fast + efficient



Evaluation Methodology

Evaluation Criteria

- Answer Relevancy
- Use of Correct Law / Section
- Consistency with retrieved text
- Consistency with the gold answers



Demo

Results & Comparison

1. Observations:

- Accuracy (vs gold answer)
 - Llama 3.2: 0.736
 - Phi-3: 0.702
 - Mistral: 0.802 (highest)
- Faithfulness (vs retrieved context)
 - Llama 3.2: 0.637
 - Phi-3: 0.664 (highest)
 - Mistral: 0.642

Model	Accuracy	Faithfulness	Detail	Hallucination	Overall	OverallScore
Mistral	★★★★★	★★★	★★★	Medium	Best	0.722283
Llama 3.2	★★★★★	★★★	★★★	Medium	Good	0.687260
Phi-3	★★★★★	★★★	★★★★★	Medium	Weakest	0.683102

Results & Comparison

	Response Similarity with the Gold Answer			Response Similarity with the Retrieved Context			Response Length		
Questions	Llama Accuracy	Phi Accuracy	Mistral Accuracy	Llama Faithfulness	Phi Faithfulness	Mistral Faithfulness	Llama Detail	Phi Detail	Mistral Detail
1	76.69%	75.70%	89.66%	71.69%	82.28%	75.88%	33	42	89
2	83.04%	53.89%	85.94%	55.38%	50.71%	53.74%	49	46	38
3	80.05%	71.31%	86.22%	70.84%	71.86%	74.26%	34	161	54
4	91.79%	52.54%	96.26%	54.28%	66.82%	52.58%	9	462	10
5	75.76%	66.02%	66.22%	61.81%	56.72%	58.69%	10	28	13
6	86.08%	71.21%	80.92%	71.10%	71.58%	64.17%	21	41	26
7	70.05%	63.17%	74.13%	68.63%	67.31%	66.34%	30	60	27
8	84.30%	74.50%	82.50%	69.19%	67.02%	66.80%	24	46	34
9	65.81%	76.40%	71.93%	63.26%	62.93%	70.02%	14	56	32
10	76.65%	87.15%	86.34%	62.69%	61.50%	56.71%	35	49	35
11	55.13%	70.88%	72.93%	64.97%	68.96%	66.88%	44	73	100
12	51.39%	76.86%	55.25%	52.49%	73.71%	59.16%	7	55	17
13	60.35%	66.86%	89.50%	67.81%	58.63%	61.86%	46	58	18
14	74.38%	76.02%	84.95%	58.73%	70.12%	72.55%	20	51	45

Conclusion & Future Work

Key Takeaways:

- RAG significantly improves factual accuracy
- Retrieval grounding is essential for rule-based questions
- Larger models perform better but are slower

Future Enhancements:

- Multi-retriever fusion
- Better chunking strategy
- Fine-tuning on cricket-specific data
- Deploy as a web app (Streamlit / FastAPI)





**THANK
YOU**

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